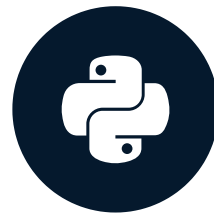


Lists in Python

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Instructor

Lists - square brackets []

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

Python is zero-indexed

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

Index:

0

1

2

3

4

5

Subset lists

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

```
months[0]
```

```
'January'
```

```
months[2]
```

```
'March'
```

Negative indexing of lists

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

```
months[-1]
```

```
'June'
```

```
months[-2]
```

```
'May'
```

Subsetting multiple list elements with slicing

Slicing syntax

```
# Includes the start and up to (but not including) the end  
mylist[startAt:endBefore]
```

Example

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

```
months[2:5]
```

```
['March', 'April', 'May']
```

```
months[-4:-1]
```

```
['March', 'April', 'May']
```

Extended slicing with lists

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

```
months[3:]
```

```
['April', 'May', 'June']
```

```
months[:3]
```

```
['January', 'February', 'March']
```


Slicing with Steps

```
# Includes the start and up to (but not including) the end  
mylist[startAt:endBefore:step]
```

```
months = ['January', 'February', 'March', 'April', 'May', 'June']
```

```
months[0:6:2]
```

```
['January', 'March', 'May']
```

```
months[0:6:3]
```

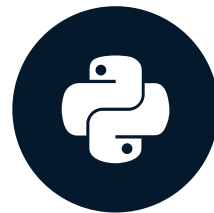
```
['January', 'April']
```


Let's practice!

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Lists in Lists

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Lists in Lists

- Lists can contain various data types, including lists themselves.
- Example: a nested list describing the month and its associated consumer price index

```
cpi = [['Jan', 'Feb', 'Mar'], [238.11, 237.81, 238.91]]
```

Subsetting Nested Lists

```
months = ['Jan', 'Feb', 'Mar']  
print(months[1])
```

```
'Feb'
```

```
cpi = [['Jan', 'Feb', 'Mar'], [238.11, 237.81, 238.91]]  
print(cpi[1])
```

```
[238.11, 237.81, 238.91]
```

More on Subsetting Nested Lists

How would one subset out a specific price index?

```
cpi = [['Jan', 'Feb', 'Mar'], [238.11, 237.81, 238.91]]  
print(cpi[1])
```

```
[238.11, 237.81, 238.91]
```

```
print(cpi[1][0])
```

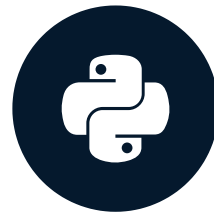
```
238.11
```

Let's practice!

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Methods and functions

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Methods vs. Functions

Methods

- All methods are functions
- List methods are a **subset** of built-in functions in Python
- Used on an object
 - `prices.sort()`

Functions

- Not all functions are methods
- Requires an input of an object
 - `type(prices)`

List Methods - sort

- Lists have several built-in methods that can help retrieve and manipulate data
- Methods can be accessed as `list.method()`

`list.sort()` sorts list elements in ascending order

```
prices = [238.11, 237.81, 238.91]
prices.sort()
print(prices)
```

```
[237.81, 238.11, 238.91]
```

Adding to a list with append and extend

`list.append()` adds a single element to a list

```
months = ['January', 'February', 'March']  
months.append('April')  
print(months)
```

```
['January', 'February', 'March', 'April']
```

`list.extend()` adds each element to a list

```
months.extend(['May', 'June', 'July'])  
print(months)
```

```
['January', 'February', 'March', 'April', 'May', 'June', 'July']
```

Useful list methods - index

`list.index(x)` returns the lowest index where the element x appears

```
months = ['January', 'February', 'March']  
prices = [238.11, 237.81, 238.91]
```

```
months.index('February')
```

```
1
```

```
print(prices[1])
```

```
237.81
```

More functions ...

- `min(list)` : returns the smallest element
- `max(list)` : returns the largest element

Find the month with smallest CPI

```
months = ['January', 'February', 'March']  
prices = [238.11, 237.81, 238.91]
```

```
# Identify min price  
min_price = min(prices)  
# Identify min price index  
min_index = prices.index(min_price)  
# Identify the month with min price  
min_month = months[min_index]  
print(min_month)
```

February

Let's practice!

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